A Retail Digital Rupiah Prototype on Hyperledger Fabric: Two-Tier Business Processes and a Reproducible Benchmark

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***Abstract*—Bank Indonesia's public Project Garuda experiments focus on wholesale central bank digital currency. This paper asks how a testable retail Digital Rupiah prototype can represent a two-tier distribution model and institutional oversight on a permissioned ledger. We implement the prototype with Hyperledger Fabric 2.5, Go chaincode and backend services, CouchDB, and PostgreSQL. The prototype separates institutional and retail roles. Bank Indonesia issues Digital Rupiah into Treasury and distributes it directly to a validator bank or payment service provider acting as custodian. Backend ownership checks and chaincode MSP checks enforce user and organization boundaries, while the shared Fabric channel does not provide transaction privacy. Automated tests evaluate authorization and transfer safety. Hyperledger Caliper uses isolated transaction types, a fixed-rate pilot, calibration at one, two, four, and eight workers, and five primary repetitions per load. Calibration selects one worker for the primary study. Under the declared sustainability gate of at least 99% successful completions and successful-transaction p95 latency at or below 2.5 seconds in every repetition, sustainable loads reach 190 TPS for balance queries, 50 TPS for wallet freeze, 65 TPS for wallet unfreeze, 25 TPS for redemption, and 20 TPS for low-contention retail transfer and merchant payment. Issuance, distribution, and deliberate high-contention retail transfer do not meet the gate at the tested rates. A separate five-repetition Go/PostgreSQL/KYC demonstration evaluates the service path. These measurements describe a single-host prototype and do not establish production capacity or cross-machine reproducibility.**

***Index Terms*—Digital Rupiah, retail CBDC, Hyperledger Fabric, two-tier distribution, performance evaluation**

I. Introduction

Central banks are evaluating Central Bank Digital Currency (CBDC) as a digital form of sovereign money [1]. Bank Indonesia pursues CBDC through Project Garuda. Its publicly documented proof of concept targets the *wholesale* tier, specifically a cash ledger for issuance and redemption among the central bank and large financial institutions [2], [3]. The Garuda White Paper also describes a retail tier in which intermediary banks and payment service providers (PJP) distribute Digital Rupiah to households and merchants [3]. That description remains at the policy and architecture level, and we are not aware of an openly available, testable system that implements Bank Indonesia's retail money business processes end to end on a blockchain ledger.

Central-bank digital money involves a sequence of business processes, not a single ledger operation. The central bank *issues* money into its Treasury and *distributes* liquidity directly to a validator bank or PJP custodian. Custodians serve customers and merchants in the retail tier. Holders *transact*, custodians *redeem* money to the central bank, and the supervisor *reconciles* and audits its circulation. A blockchain implementation must express these processes as deterministic transactions that each organization can validate independently. It must also preserve two-tier intermediation to reduce bank disintermediation [4] and give the supervisor an audit surface.

Most technical studies isolate one part of this problem: permissioned CBDC architecture [5], privacy [6], security threat models [7], or throughput in account-based systems [8]. Deployed retail systems, including Nigeria's eNaira [9], China's e-CNY [10], and the Bahamian Sand Dollar [11], publish design papers but not reproducible performance benchmarks. The surveyed literature therefore provides no integrated, openly measurable retail reference for the Indonesian two-tier model [12], [13].

To address this gap, we implement a Digital Rupiah prototype on a permissioned Hyperledger Fabric network [14]. The prototype maps Bank Indonesia's business processes to chaincode transactions. This paper contributes:

1) A two-tier Digital Rupiah chaincode that integrates hierarchical issuance and distribution, retail transfers, and supervisor-facing reconciliation. This extends the wholesale-only Garuda proof of concept with a testable retail implementation.
2) A deterministic transfer protocol with explicit authorization and balance checks, Fabric multi-version concurrency control (MVCC), and atomic wallet updates. The protocol lets endorsing organizations apply the same state-transition rules independently.

3) An open, reproducible retail-CBDC benchmark with Hyperledger Caliper [15] that documents the testbed, workload, deterministic population seeding, offered-load profiles, and authorization and aggregate-overspend checks. The benchmark reports throughput, latency, success rate, and container-level resource measurements while keeping the local testbed and one-worker scope explicit.

II. Background and Related Work

A. Two-Tier Retail CBDC

In the two-tier (intermediated) model the central bank issues CBDC and remains the ultimate liability holder, while regulated intermediaries serve end users and operate customer-facing rails [1], [4]. This preserves the existing division of labor and limits the central bank's exposure to retail operations. Bank Indonesia adopts this structure for Digital Rupiah, with banks and PJP acting as distributors between the central bank and the public [3].

B. Permissioned Blockchain and Hyperledger Fabric

Hyperledger Fabric is a modular, permissioned blockchain with pluggable ordering, channel-based data isolation, and separate execution, ordering, and validation stages [14]. Membership Service Providers (MSPs) govern identity, the crash-fault-tolerant Raft protocol orders transactions, and CouchDB can provide a queryable world state. Fabric has consequently been used in CBDC and regulated-finance prototypes [5], [16]. Hyperledger Caliper [15], [17] measures throughput and latency under different workloads.

C. Authorization and Supervision

Retail CBDC systems require explicit authorization boundaries, safe balance updates, and supervisory visibility. Token-based retail designs also face denial-of-service and double-spend risks [18], so transfer validation must be deterministic. Our chaincode combines organization-aware authorization, wallet-state checks, and supervision events, complementing studies that address privacy [6] or security [7] separately. The closest published prototype is Project Aurum, a full-stack two-tier retail CBDC built for the Hong Kong Monetary Authority [19], which pairs a wholesale interbank system with a retail e-wallet system and, like this work, keeps the two tiers architecturally distinct. Aurum reports no reproducible throughput benchmark, and its source code and technical manuals are released only to BIS member central banks rather than publicly. What distinguishes this prototype among the representative systems in Table I is that its two-tier intermediation and authorization and supervision controls are realized in a publicly available implementation with a reproducible benchmark, rather than described in design documentation alone.

TABLE I
Comparison with Representative Retail CBDC Systems

System	Platform	2-tier	Open	Bench.
e-CNY [10]	Permissioned	✓	×	×
eNaira [9]	Fabric	✓	×	×
Sand Dollar [11]	Permissioned	✓	×	×
OpenCBDC [8]	Account	×	✓	✓
This work	Fabric 2.5	✓	✓	✓

Open: implementation publicly available. Bench.: open, reproducible throughput/latency benchmark. OpenCBDC is account-based and not distributed-ledger-based. Rows span deployed national CBDCs and a research platform, and summarize public information rather than a controlled comparison.

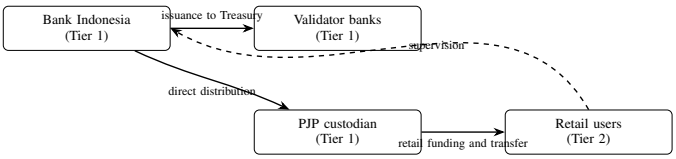


Fig. 1. Two-tier actor structure of the Digital Rupiah prototype. Bank Indonesia issues Digital Rupiah into Treasury and distributes it directly to a validator bank or PJP custodian, which serves retail users. Dashed flow denotes supervisory visibility.

TABLE II
Bank Indonesia Business Processes and Their Chaincode Realization

Process	Initiator	Chaincode transaction
Issuance	Bank Indonesia	`Mint`, `RequestIssuance`
Distribution	Bank Indonesia	`DistributeToParticipant`
Wallet creation	Bank / PJP	`CreateWallet`
Payment	Customer	`Transfer`
Redemption	Participant	`Burn`, `RequestRedemption`
Supervision	OJK / BI	`GetReconciliationReport`, `GetMetrics`, `GetSupervisionEvents`

The channel endorsement policy accepts an institutional MSP from Bank Indonesia, HimbaraBank, CommercialBank, or PJP; OJK is observer-only.

III. Bank Indonesia Business Processes

The system implements six processes in the circulation of central-bank digital money. Fig. 1 shows the actors in each tier. Table II identifies the initiator, chaincode transaction, and endorsement requirement for each process.

The institutional tier contains Bank Indonesia, validator banks, and PJP Custodians. These participants follow a lifecycle of submission, approval, freezing, unfreezing, rejection, and offboarding. Bank Indonesia mints Digital Rupiah only to the Treasury wallet, then distributes liquidity directly to an active validator bank or PJP through `DistributeToParticipant`. Each Custodian serves customers and merchants in the retail tier, where the shared transfer path applies ownership, wallet-status, authorization, and balance checks.

IV. System Architecture

Fig. 2 shows the four-layer architecture: client, backend API, off-chain store, and Fabric network. The web client calls

TABLE III
NETWORK TOPOLOGY OF THE FIVE-ORGANIZATION FABRIC NETWORK

Organization	Role	Peer	CouchDB
BankIndonesiaOrg	Issuer	peer0.bi:7051	:5984
HimbaraBankOrg	Validator	peer0.himbara:9051	:6984
CommercialBankOrg	Validator	peer0.commercial:11051	:7984
PJPOrg	Distributor	peer0.pjp:13051	:9984
OJKObserverOrg	Supervisor	peer0.ojk:12051	:8984
Ordering: orderer.paynet:7050 (single-node Raft); per-org CA; TLS enabled			

TABLE IV
CHAINCODE FUNCTION GROUPS

Group	Representative functions
Participants	`SubmitParticipant`, `ApproveParticipant`, `Freeze/Unfreeze`, `Offboard`
Wallets	`CreateWallet`, `CreateWholesaleWallet`, `Freeze/UnfreezeWallet`
Issuance	`Mint`, `RequestIssuance`, `RequestIssuanceRtgs`
Redemption	`Burn`, `RequestRedemption`
Distribution	`DistributeToParticipant`
Transfer	`Transfer`
Supervision	`GetReconciliationReport`, `GetMetrics`, `GetSupervisionEvents`

a Go REST backend built with Gorilla Mux. The backend authenticates requests with JWT and role-based access control, stores account and authentication data in PostgreSQL, and submits ledger transactions through the Fabric Gateway SDK. Fig. 3 details the permissioned network.

A. Network Topology

The network runs five peer organizations on one application channel (Fig. 3, Table III). *BankIndonesiaOrg* is the issuer; *HimbaraBankOrg* (HIMBARA, Indonesia's association of state-owned banks) and *CommercialBankOrg* are validator and distributor banks; *PJPOrg* represents the payment service provider; and *OJKObserverOrg* is the supervisor (OJK, *Otoritas Jasa Keuangan*, the financial-services authority), which initiates no business transactions and is excluded from chaincode approval. Each organization operates one peer with a CouchDB world state and maintains its own certificate authority and MSP for identity management. A single Raft orderer orders transactions; this configuration provides ordering but not crash fault tolerance, which requires at least three consenters. The channel endorsement policy accepts one of the four institutional MSPs, while the chaincode lifecycle policy requires Bank Indonesia and one institutional participant.

B. On-Chain and Off-Chain Data

The Go chaincode in `digital_rupiah.go` groups functions for participants, wallets, issuance and redemption, retail transfers, and supervision (Table IV). Account and authentication data remain in the off-chain PostgreSQL store, while the ledger contains the wallet and transaction state needed for settlement and oversight.

TABLE V
BACKEND ROLES AND REPRESENTATIVE PERMISSIONS

Role	Representative permitted actions
Public	Health check, login
Authenticated	View own profile, list wallets
Bank/PJP	Create wallet, fund retail wallets
Bank Indonesia	Issue, redeem liquidity, approve participant
Merchant	Initiate transfer, view balance
Supervisor	Reconciliation, metrics, supervision events

A Go backend enforces role-based access control over six roles, summarized in Table V, before any transaction reaches the chaincode.

The two layers enforce different things, and the split matters for how the prototype's guarantees should be read. The backend maps JWT roles to permitted operations and restricts customers to their own wallets. The chaincode checks the caller's organization, wallet status, transfer amount, and available balance before it writes the state transition. Every endorsing peer re-evaluates these rules independently, while the channel endorsement policy restricts validation to institutional MSPs. The single backend and PostgreSQL instance remain trust and availability concentration points for this prototype.

V. PROTOCOL AND DETERMINISM

The transfer path applies explicit authorization and balance checks during endorsement. We formalize these checks and examine their behavior under concurrent transactions.

A. Transfer-Validity Rules

For a transfer of amount a from sender s to receiver r, the chaincode admits the transfer only if every one of the following checks passes:

- the caller's organization and wallet ownership satisfy the authorization rules;
- neither the sender's nor the receiver's wallet is frozen;
- $a > 0$ and the required business reference is present;
- the sender has enough balance to cover a; and
- the sender and receiver wallet records can be updated atomically.

If any check fails, the transfer is rejected. A successful transfer decreases the sender balance, increases the receiver balance, records the transaction, and emits the supervision event used by the read-only observer path. Treasury distribution accepts only active validator-bank or PJP custodians, and a PJP cannot call issuance directly.

B. Determinism and Double-Spend Safety

Every endorsing peer evaluates the authorization and transfer rules against the same transaction arguments and wallet records. The state transition does not depend on a peer-local clock or on a peer-local cache. Each successful transfer updates the sender and receiver records in one transaction, so every endorser evaluates the same balance transition before validation.

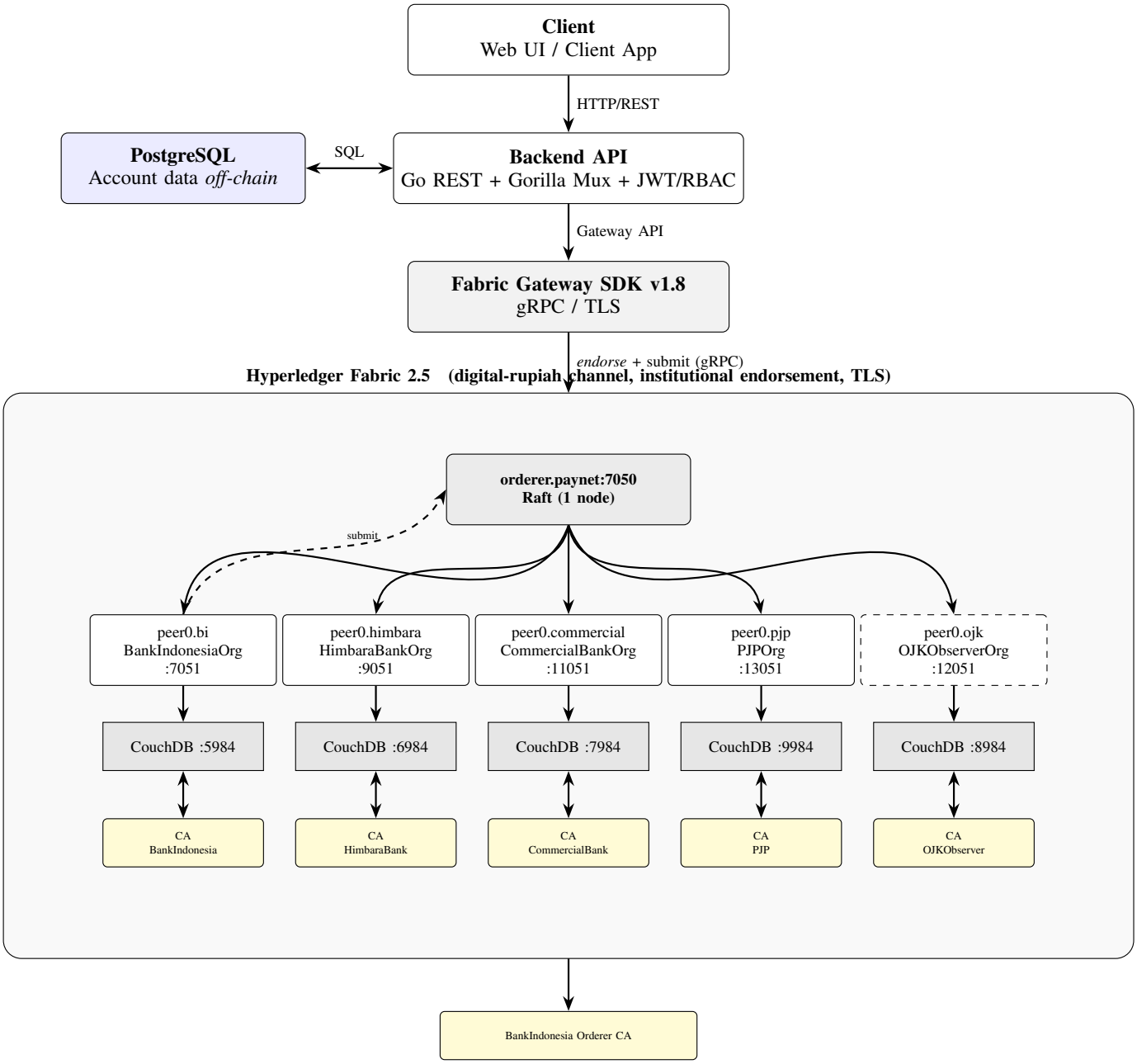


Fig. 2. Full system architecture of the Digital Rupiah prototype: client layer, Go REST backend with JWT/RBAC, off-chain PostgreSQL store, Fabric Gateway SDK, and the Hyperledger Fabric 2.5 network of five peer organizations with CouchDB, a per-organization CA, and a Raft orderer.

Double-spend safety follows from Fabric's multi-version concurrency control (MVCC). An endorsement reads the current version of each wallet record, and Fabric commits the transaction only if those versions remain unchanged through ordering and validation. Two concurrent transfers from the same wallet may endorse against the same record and version, but only the first ordered transaction commits. The second fails MVCC validation after the first update. Because the balance records are the keys in the read set, one version check covers the state used by the transfer and prevents a stale balance update.

VI. Performance Evaluation

A. Methodology

We evaluate the final prototype with Hyperledger Caliper 0.6.0 on a clean ledger for each profile. The combined manifest `20260909-final-evidence-manifest.json` records suite seed 20260909 (profile ordering); the transfer configurations retain workload seed 20260725. It also records chaincode `digital-rupiah` version 3.0 sequence 1, runtime versions, input hashes, result hashes, and the single-host topology. The testbed has five peer organizations, one

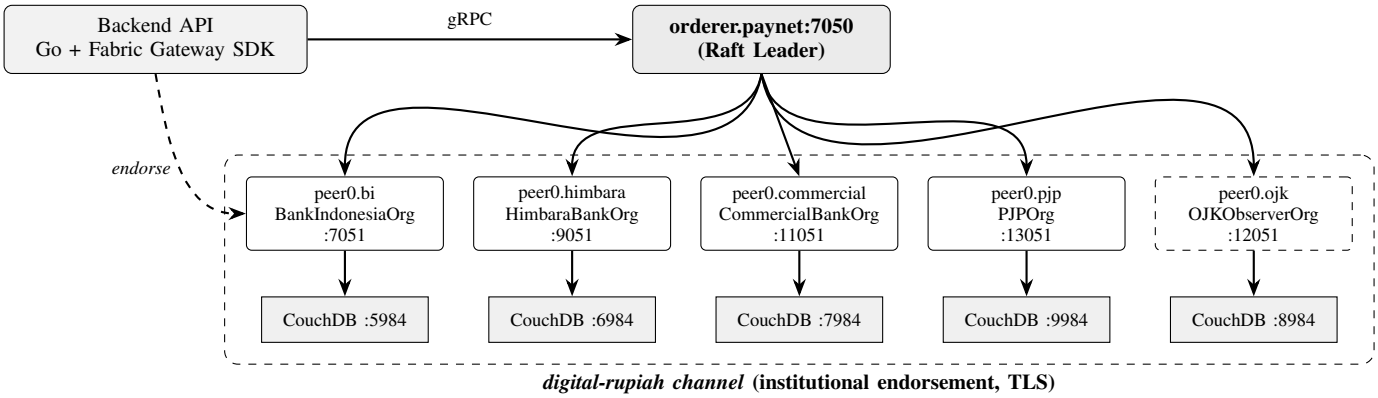


Fig. 3. Hyperledger Fabric network topology: five peer organizations, each with a CouchDB world state, on a single channel under a Raft orderer; the Go backend endorses via the Fabric Gateway SDK.

```
function TRANSFER(sender, receiver, amount)
if caller unauthorized or either wallet is frozen then
    return REJECT
end if
if amount ≤ 0 or reference is missing then
    return REJECT
end if
if sender.balance < amount then
    return REJECT
end if
sender.balance ← sender.balance − amount
receiver.balance ← receiver.balance + amount
append transaction history and emit supervision event
PutState (validated by Fabric MVCC)
return COMMIT
```

Fig. 4. Authorized, atomic, and double-spend-safe transfer executed by the chaincode. Fabric MVCC validates the wallet versions before commit.

TABLE VI
FINAL TESTBED CONFIGURATION

Component	Specification
Processor	Intel Core i7-7700HQ, 4 cores, 8 threads, 2.80 GHz
Memory	15.52 GiB
Runtime	Node.js 22.23.2, npm 10.9.1, Docker 27.5.1
Fabric / CouchDB	2.5 / 3.2
Caliper	0.6.0
Chaincode	`digital-rupiah` v3.0, sequence 1
Ordering	Single-node Raft; batch timeout 2 s; maximum 10 messages
Endorsement	One institutional MSP: BI, HimbaraBank, CommercialBank, or PJP

peer per organization, one Raft orderer, and one CouchDB instance per peer.

The primary study isolates each transaction type and contention mode. A pilot uses fixed-rate retail transfers, calibration compares one, two, four, and eight workers, and the primary study fixes the smallest worker count sustaining the highest passing load. Each primary cell has five repetitions. Offered TPS is submitted transactions divided by the measured window; achieved TPS is successful completions divided by that window. Latency is the Caliper Fabric boundary, `time_final - time_create`, for successful completions. The study also records failed and timed-out completions, failure categories, and host, container, disk, network, and Caliper-client resource metrics.

TABLE VII
SUSTAINABLE PRIMARY LOADS BY TRANSACTION TYPE

Function	Contention	Load	Offered (TPS)	Achieved (TPS)	Success (%)	p95 (s)	p99 (s)
Balance query	Low	190	190.0	190.0	100.00	0.01	0.01
Wallet freeze	Low	50	50.0	50.0	100.00	0.28	0.37
Wallet unfreeze	Low	65	65.0	65.0	100.00	0.27	0.35
Redemption	Low	25	25.0	25.0	100.00	0.53	0.60
Retail transfer	Low	20	20.0	20.0	100.00	0.72	0.89
Merchant payment	Low	20	20.0	20.0	100.00	0.66	0.75
Issuance	Low	–	–	–	<99	–	–
Distribution	Low	–	–	–	<99	–	–
Retail transfer	High	–	–	–	<99	–	–

TABLE VIII
FAILURE CATEGORIES IN PRIMARY WORKLOADS

Category	Count
MVCC read conflict	358,413
Unknown failure	1,013,243

The sustainability rule was fixed before interpretation: every one of five repetitions had to reach at least 99% successful completions and successful- transaction p95 latency at or below 2.5 seconds. The table reports the highest passing load for each operation. Issuance, distribution, and high-contention retail transfer have no passing load in the tested range; their failed submissions are retained in the failure analysis rather than converted into a throughput claim.

The primary logs classify 358,413 failures as MVCC read conflicts and 1,013,243 as Unknown failure. Status 11 indicates that a transaction read a state version that changed before validation. Unknown failure is retained when the available log

TABLE IX
FIVE-REPETITION SUMMARY AT THE HIGHEST PASSING LOADS

Function	Load	Offered	Achieved	Success (%)	p95 (s)
Balance query	190	190.0	190.0	100.00	0.01
Wallet freeze	50	50.0	50.0	100.00	0.28
Wallet unfreeze	65	65.0	65.0	100.00	0.27
Redemption	25	25.0	25.0	100.00	0.53
Retail transfer	20	20.0	20.0	100.00	0.72
Merchant payment	20	20.0	20.0	100.00	0.66

TABLE X
SEPARATE POSTGRESQL/KYC DEMONSTRATION

Stage	Runs	Mean (ms)	SD (ms)
Login	5	186.85	34.77
Customer creation	5	2,069.02	8.60
KYC submission	5	2,071.99	7.02
KYC refresh	5	2,064.70	8.31
Wallet creation	5	2,075.16	6.05
KYC read	5	1.14	0.41

TABLE XI
SEMANTIC CONFORMANCE PROFILES

Profile	Succ.	Fail	Oracle result
Authorization	2	1	PASS; one expected custodian-mismatch rejection.
Boundary	4	4	PASS; 8/8 cases, zero mismatches, zero infrastructure errors.
Negative path	0	7	PASS; 7/7 expected rejection rules enforced.
Aggregate overspend	10	190	PASS; final sender balance Rp40,000.

does not support a more specific category; it is not interpreted as proof of a single bottleneck.

Each row is the mean of five repetitions. The complete per-run standard deviations, p50, p95, p99, minimum, and maximum values remain available in the machine-readable benchmark summary. The table excludes operations without a passing load under the declared sustainability rule.

This five-repetition demonstration measures the Go/PostgreSQL/KYC service path separately from the direct Fabric benchmark. It covers login, customer creation, KYC submission and refresh, wallet creation, and KYC reads. Its latencies must not be combined with the Fabric transaction latency boundary.

The failed counts in these semantic profiles represent expected policy rejections. The aggregate-overspend workload requested Rp200,000 from an initial Rp50,000 sender balance. Ten transfers committed and 190 were rejected; the final balance remained Rp40,000. These observations support the tested authorization and balance invariants, but do not constitute a formal security proof.

The resource sampler records host, container, disk, network, Prometheus, and Caliper-client measurements. The primary results are interpreted together with these measurements and the categorized failures; they remain bounded to the single-host testbed and do not establish production capacity.

TABLE XII
RESOURCE METRICS COLLECTED IN THE PRIMARY STUDY

Layer	Metrics
Host	CPU, memory, I/O, IOPS, and disk latency
Fabric containers	CPU, memory, network, IOPS, and disk service latency
Caliper client	Process CPU
Prometheus	Monitoring CPU and memory

VII. LIMITATIONS AND FUTURE WORK

The evaluation uses a single host with one peer per organization and no geographic distribution or network latency. Its Caliper results therefore apply only to the reported configuration and do not estimate national-scale performance. Each primary profile uses a clean ledger, a fixed seeded population, one calibrated worker, and five repetitions. A distributed deployment and a graded-skew contention workload would be needed to characterize other operating conditions.

The shared Fabric channel does not provide transaction privacy, and the prototype does not use private data collections or privacy-preserving cryptography. The single orderer provides ordering but does not tolerate its own failure. The prototype also omits offline payment and leaves connections to domestic infrastructure such as BI-RTGS and BI-FAST as stubs.

The authorization and aggregate-overspend profiles cover the tested paths only. They do not establish formal security, production readiness, or cross-machine reproducibility. Future work should repeat the protocol on distributed hosts, instrument endorsement and commit phases, expand the contention workload, and evaluate privacy, offline payment, and interoperability.

VIII. CONCLUSION

This paper implements Bank Indonesia's digital money business processes in a two-tier Digital Rupiah prototype on a five-organization Hyperledger Fabric network. The chaincode integrates hierarchical issuance and distribution, retail transfers, organization-aware authorization, and supervisor-facing reconciliation.

The transfer protocol combines explicit wallet-state checks with Fabric MVCC. Under the declared sustainability gate, the primary study reaches 190 TPS for balance queries, 50 TPS for wallet freeze, 65 TPS for wallet unfreeze, 25 TPS for redemption, and 20 TPS for low-contention retail transfer and merchant payment. Issuance, distribution, and deliberate high-contention retail transfer do not meet the gate at the tested rates. All primary cells use five repetitions, and the final report separates offered TPS, achieved TPS, failure categories, latency percentiles, resource measurements, and variability. These results provide a local, auditable baseline for retail Digital Rupiah experiments, not a production-capacity claim.

ARTIFACT AVAILABILITY

The chaincode, the five-organization network definition, the Go backend, and the Hyperledger Caliper workloads and benchmark configurations used in this paper are available at https://github.com/mfachrizalg/bi-coin.

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